Exp No:1.a Analyse the trend of data science job postings over the last

Decade

CODE

```
import pandas as pd

import matplotlib.pyplot as plt

data = {'Year': list(range(2010, 2021)),

'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}

df = pd.DataFrame(data)

plt.plot(df['Year'], df['Job Postings'], marker='o')

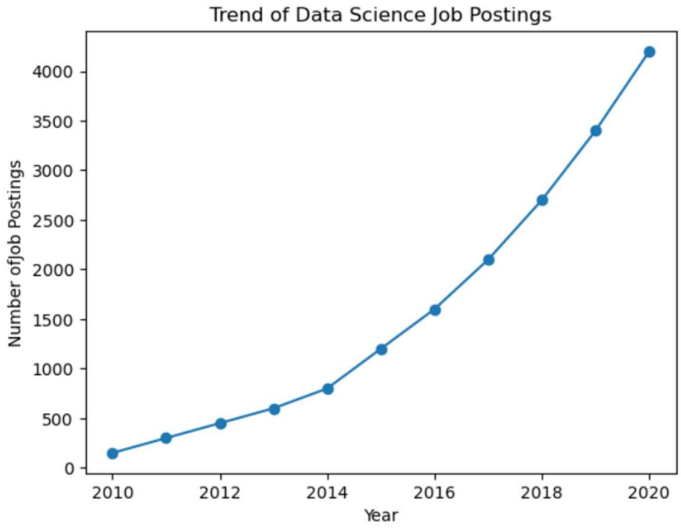
plt.title('Trend of Data Science Job Postings')

plt.xlabel('Year')

plt.ylabel('Number ofJob Postings')

plt.show()
```

output



Result: the trend of data science job postings for the last decade has been analyzed

Exp No:1.b Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Code

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',

'Business Analyst']

counts = [300, 500, 450, 200, 150]

plt.bar(roles, counts)

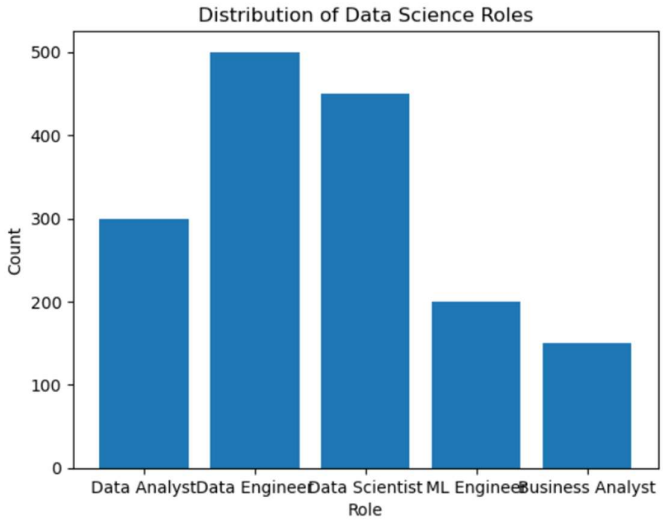
plt.title('Distribution of Data Science Roles')

plt.xlabel('Role')

plt.ylabel('Count')

plt.show()
```

output



Result: the distribution of various data science roles has been analysed

Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semi-structured data based on data sets given.

CODE:

```
structured_data = pd.DataFrame({

'ID': [1, 2, 3],

'Name': ['Sally Madhulika', 'Sanjita', 'Sangavi'],

'Age': [25, 30, 35]

})

print("Structured Data:\n", structured_data)

unstructured_data = "my name is Sally Madhulika”

print("\nUnstructured Data:\n", unstructured_data)

semi_structured_data = {'ID': 1, 'Name': 'Sally Madhulika’,
'Attributes': {'Height': 170, 'Weight': 60}}

print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT:

```
Structured Data:
    ID             Name  Age
0   1  Sally Madhulika   25
1   2          Sanjita   30
2   3          Sangavi   35

Unstructured Data:
 My name is Sally Madhulika

Semi-structured Data:
 {'ID': 1, 'Name': 'Sally Madhulika', 'Attributes': {'Height': 170, 'Weight': 60}}
```

RESULT: hence, Structured , Un-structured and Semi-structured data are differentiated.

Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data.

CODE:

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()

f = Fernet(key)

token = f.encrypt(b"Rajalakshmi Engineering College")

token

b'...'

f.decrypt(token)

b'Rajalakshmi Engineering College'

key = Fernet.generate_key()

cipher_suite = Fernet(key)

plain_text = b"Rajalakshmi Engineering College."

cipher_text = cipher_suite.encrypt(plain_text)

decrypted_text = cipher_suite.decrypt(cipher_text)

print("Original Data:", plain_text)

print("Encrypted Data:", cipher_text)

print("Decrypted Data:", decrypted_text)
```

OUTPUT:

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABqeWlqKrpskFqHBbhirNcJIdQBRWYG1yD8Mub-tmK5MoReCIb12-ME25PIv-cZIxPyhUfJvm8C9HM1tHAi2lWGyoXYLHx9H7GSaOP4DPfpnZMnYwfT4btsJr6w9tA4omG_1egq'
Decrypted Data: b'Rajalakshmi Engineering College.'
```

RESULT: Hence given sensitive data has been encrypted and decrypted.